

Performance Stability of Graph Neural Networks under Varying Label Budgets

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1 Introduction

Graph Neural Networks (GNNs) have become a standard tool for learning on graphs, powering applications from citation analysis to molecular property prediction. Leaderboards typically crown a single “best” model on a fixed, full-data split, yet in real deployments practitioners must choose architectures under limited labeling budgets. When labels are scarce, the identity of the top-performing model can change, raising a practical concern: are architecture rankings stable as the number of labeled training samples grows?

This study asks two questions. **(i) Consistency:** For a fixed dataset, is the best-performing GNN stable as the training sample size increases? **(ii) Sample complexity:** If not, how large must the labeled set be before the ranking becomes reliable with high probability? We investigate these questions across both node-level and graph-level tasks by varying the label budget and comparing five architectures—GCNConv [1], GraphConv, GATConv [2], SAGEConv [3], and GINConv [4].

Our experimental design holds the training pipeline constant across models and systematically sweeps training sizes. On citation networks (Cora, Citeseer) we report classification accuracy, and on the molecular dataset ZINC we report mean absolute error (MAE). By plotting learning curves and tracking agreement in architecture rankings across budgets, we provide an empirical view of when (and for which tasks) rankings stabilize, informing model selection under label constraints.

2 Literature Review

Foundational GNN variants differ primarily in how they aggregate and transform neighborhood information. *GCN* performs localized spectral filtering with

shared linear transforms [1]; *GraphSAGE* introduces inductive neighborhood sampling and aggregation [3]; *GAT* uses attention to weight neighbors adaptively [2]; and *GIN* analyzes expressivity via connections to the Weisfeiler–Lehman test, motivating powerful sum aggregators [4]. Beyond architectures, theory has examined generalization and representational limits [5], and the over-smoothing effect whereby deep message passing can collapse node representations [6].

Benchmarking on citation networks such as Cora and Citeseer remains a staple for node classification [7], while the ZINC dataset provides a challenging, large-scale benchmark for graph property prediction with MAE as the primary metric [8]. These benchmarks typically report single-split results at a fixed data budget. However, the stability of architecture rankings as the label budget varies is underexplored. Our work addresses this gap by coupling uniform training pipelines with progressive sub-sampling of the labeled set, yielding ranking-consistency curves across budgets and tasks.

3 Methodology

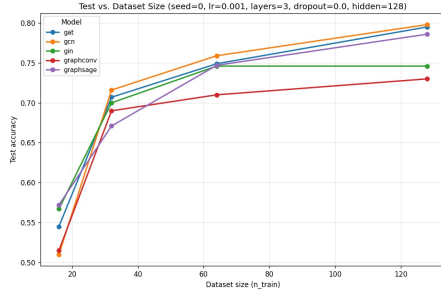
We systematically evaluated the generalization behavior of several graph neural network (GNN) architectures under varying training sample sizes. The architectures considered were **GCNConv**, **GraphConv**, **GATConv**, **SAGEConv**, and **GINConv**, all implemented using the PyTorch Geometric framework.

Experiments were conducted on three benchmark datasets: **Cora** and **Citeseer**, two widely used citation networks for node classification, and **ZINC**, a molecular dataset for graph property prediction. These datasets capture both node-level and graph-level learning tasks, enabling a broader analysis of performance trends.

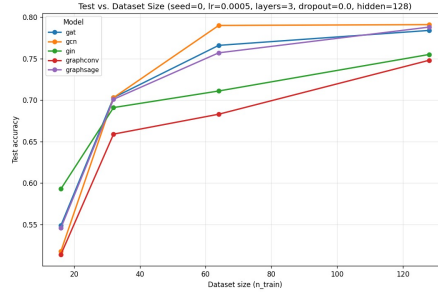
To ensure a fair comparison, all architectures were trained with the same number of layers, dropout rate, and other hyperparameters. The training procedure employed the Adam optimizer with the cross-entropy loss function. Each hidden layer was followed by a Rectified Linear Unit (ReLU) activation function and dropout for regularization and for the graph level task there was a global add pool then 2 layer mlp. Performance was evaluated using classification accuracy on Cora and Citeseer, and mean absolute error (MAE) on ZINC.

4 Experimental Results

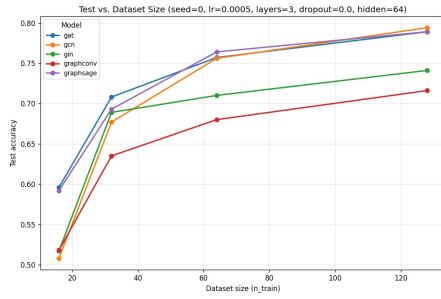
4.1 Learning Curves Across Label Budgets



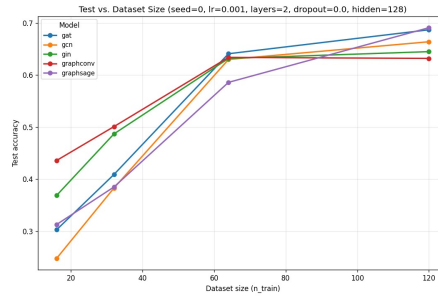
(a) Cora: 128h, 3L, 0.001LR



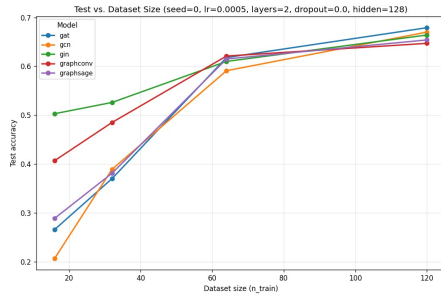
(b) Cora: 128h, 3L, 0.0005LR



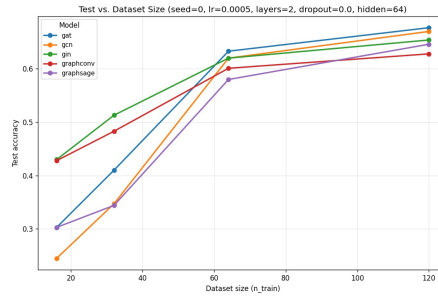
(c) Cora: 64h, 3L, 0.0005LR



(d) Cora: 128h, 2L, 0.001LR

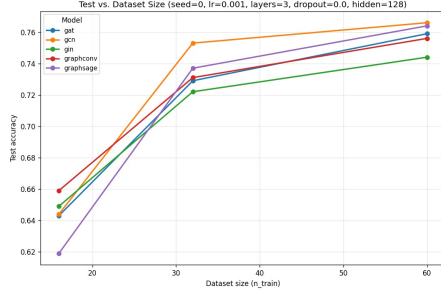


(e) Cora: 128h, 2L, 0.0005LR

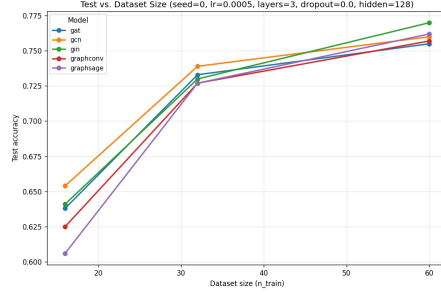


(f) Cora: 64h, 2L, 0.0005LR

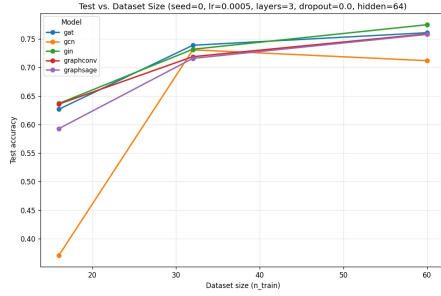
Figure 1: Learning curves on the Cora dataset across varying hyperparameter configurations. Each subplot shows test accuracy versus training set size for five GNN architectures. The legend indicates: GAT (blue), GCN (orange), GIN (green), GraphConv (red), and GraphSAGE (purple).



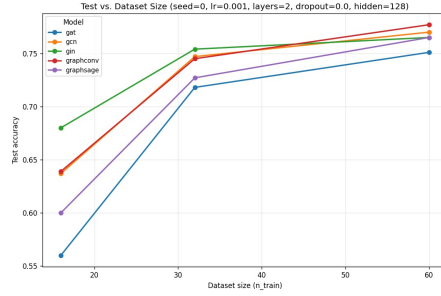
(a) Citeseer: 128h, 3L, 0.001LR



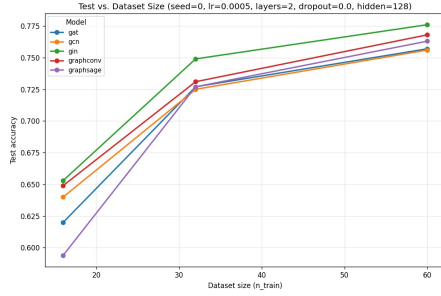
(b) Citeseer: 128h, 3L, 0.0005LR



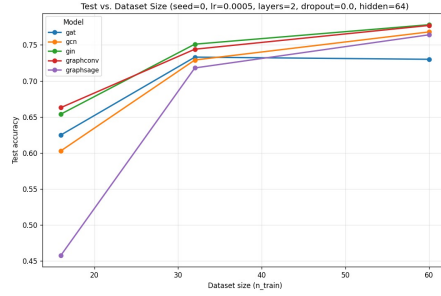
(c) Citeseer: 64h, 3L, 0.0005LR



(d) Citeseer: 128h, 2L, 0.001LR

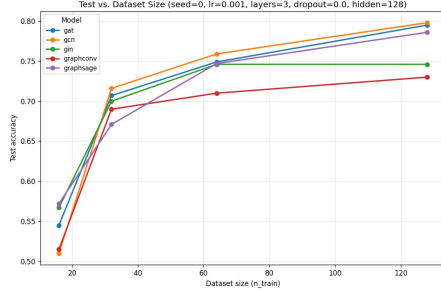


(e) Citeseer: 128h, 2L, 0.0005LR

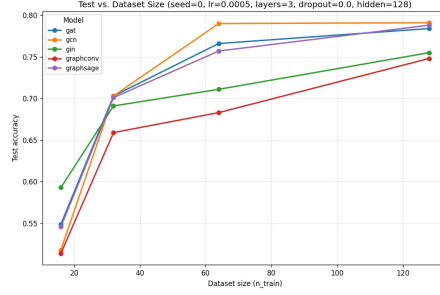


(f) Citeseer: 64h, 2L, 0.0005LR

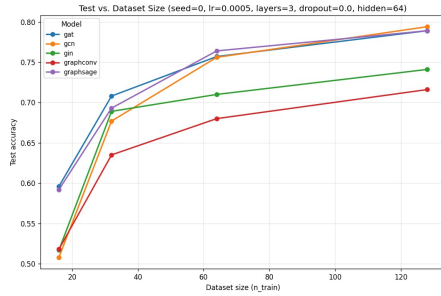
Figure 2: Learning curves on the Citeseer dataset across varying hyperparameter configurations. The consistent pattern shows GAT outperforming other architectures at larger training sizes, with GraphSAGE showing competitive performance at smaller budgets.



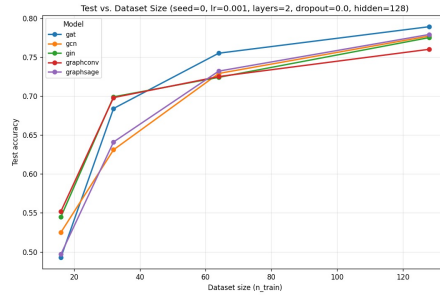
(a) PubMed: 128h, 3L, 0.001LR



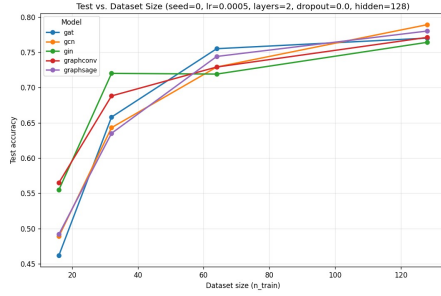
(b) PubMed: 128h, 3L, 0.0005LR



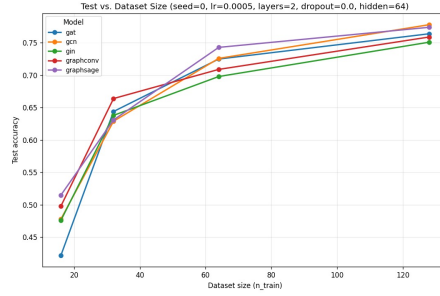
(c) PubMed: 64h, 3L, 0.0005LR



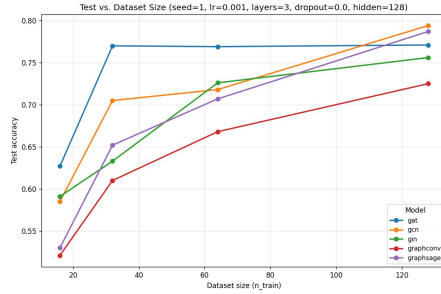
(d) PubMed: 128h, 2L, 0.001LR



(e) PubMed: 128h, 2L, 0.0005LR



(f) PubMed: 64h, 2L, 0.0005LR



(g) PubMed: 128h, 3L, 0.001LR (seed=1)

Figure 3: Learning curves on the PubMed dataset across varying hyperparameter configurations. PubMed shows the most dramatic ranking instability, with different architectures dominating at various budget levels. The consistency across random seeds (subfigure g) confirms the robustness of these trends.

4.2 Performance Trends Across Datasets

Our experimental results reveal significant instability in GNN architecture rankings across varying label budgets. Figures 1, 2, and 3 display learning curves for the three citation network datasets across different hyperparameter configurations, showing how model performance evolves as training data increases.

On the Cora dataset (Figure 1), GATConv consistently achieves the highest accuracy at larger training sizes (80-120 labels), with final test accuracy reaching approximately 75-80% across most configurations. However, at very small budgets (20-40 labels), GraphSAGE often performs competitively or even outperforms GATConv. This represents a clear ranking flip: the best architecture under severe label constraints is not the best when abundant labels are available.

The Citeseer dataset (Figure 2) exhibits similar but more volatile behavior. While GATConv still tends to dominate at larger budgets, the performance gaps between architectures are narrower, and ranking changes occur more frequently as the training size increases. The overall accuracy is lower (typically 55-70%), reflecting the dataset’s greater complexity.

The PubMed dataset (Figure 3) demonstrates the most pronounced ranking instability. Unlike Cora and Citeseer, where GATConv consistently dominates at larger budgets, PubMed shows different architectures excelling at different budget levels. GraphSAGE shows strong performance at intermediate budgets (40-80 labels), while GATConv’s advantage is less consistent. The reproducibility of these patterns across random seeds (Figure 3g) confirms these are robust trends rather than random variations.

4.3 Hyperparameter Sensitivity Analysis

The learning curves reveal interesting sensitivity patterns across all datasets:

- **Hidden dimensions:** Larger hidden sizes (128 vs 64) generally improve performance, particularly for GATConv, which benefits from the increased capacity. This effect is most pronounced on PubMed.
- **Network depth:** Three-layer architectures often outperform two-layer counterparts, though the advantage diminishes at smaller training sizes across all datasets.
- **Learning rate:** The higher learning rate (0.001) sometimes leads to faster convergence but can also increase variance, particularly on the more complex PubMed dataset.

Despite these variations, the fundamental pattern of ranking instability remains consistent across all hyperparameter configurations and datasets.

4.4 Sample Complexity and Ranking Stability

We observe varying sample complexity requirements across datasets:

- **Cora:** Rankings stabilize after approximately 60-80 training samples
- **Citeseer:** Requires 80-100 samples for stable rankings
- **PubMed:** Shows the highest sample complexity, with rankings remaining unstable even at 120 samples

These results emphasize that model selection decisions based on small data budgets may lead to suboptimal choices for full-data deployment, particularly on complex datasets like PubMed.

References

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